PHYS–467 Machine Learning for Physicists
Exercise session 9: Clustering

Exercise n. 1

1. Generate K Gaussian clusters of n samples. Each cluster has $n_k = n/K$ data points $\boldsymbol{x}^{\mu}_{i,k} \in \mathbb{R}^d$ of mean $\mu_k \in \mathbb{R}^d$, and covariance matrix $\Sigma = \sigma \mathbb{I}_{d\times d}$. Take $d = 2$;
2. Plot the data points in the 2-dimensional space with their centroids and assign to each data point a color according to cluster membership.

Exercise n. 2

1. Implement K-Means from scratch. At each iteration of the algorithm, compute the cost function and plot the detected clusters with their centroids and colors assigned according to cluster membership.
2. Plot the cost function as a function of the number of iterations.

Exercise n. 3

1. Implement K-Means using the built-in methods from scikit-learn.
2. Compare the implementation from scratch with scikit-learn.

Exercise n. 4

1. Vary the variance σ of each cluster and plot the detected cluster again with their centroids and colors assigned according to cluster membership. How does the predictions are affected when the noise level is higher?